

Zero-Shot Character Identification and Speaker Prediction in Comics via Iterative Multimodal Fusion

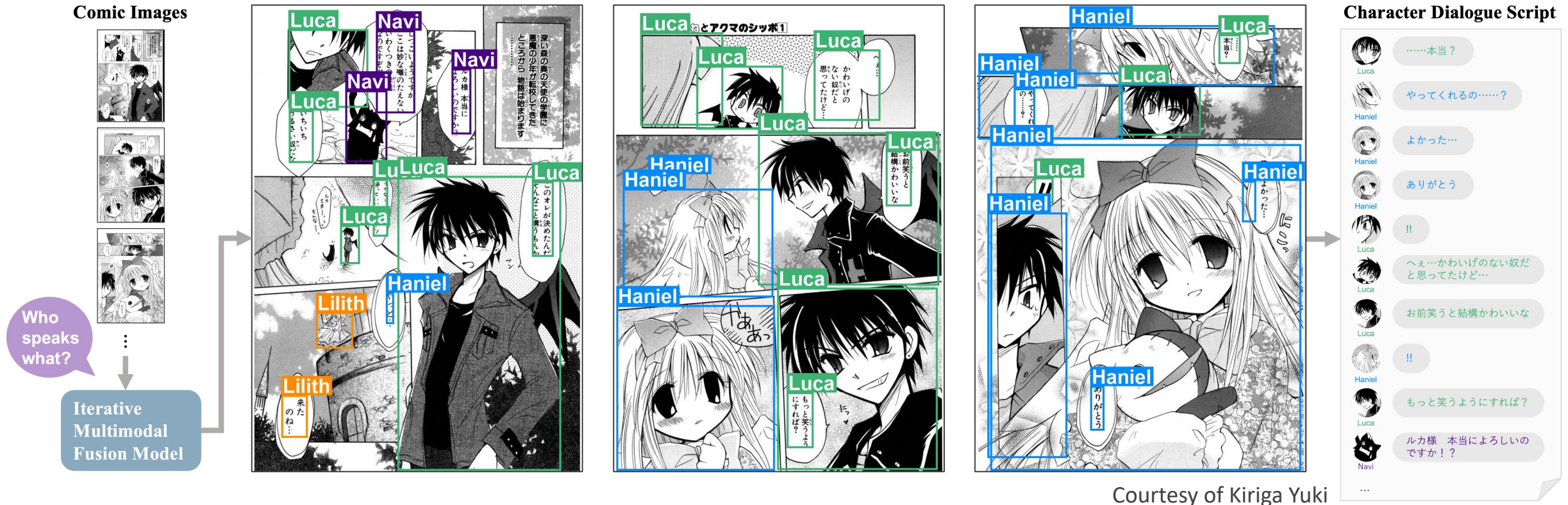
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Topic

- Zero-Shot character identification and speaker prediction in comics
 - Identify characters and predict speakers of unseen comics *only from images*



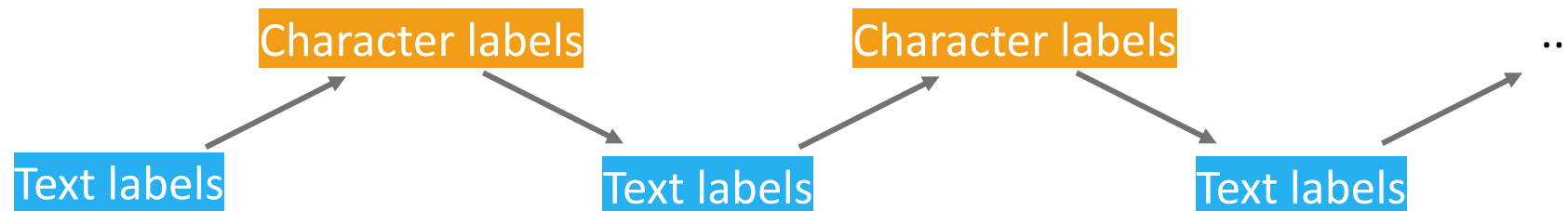
Proposal

● Research focus

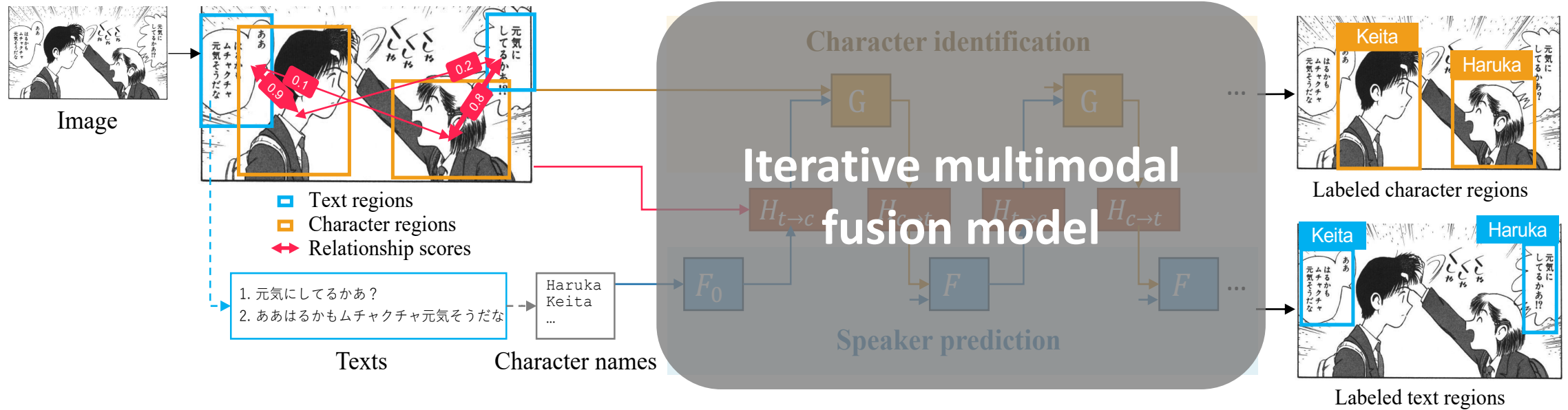
- Predict character names for both text and character regions
- Tackle **zero-shot** tasks without requiring any annotations
- Enhance real-world applicability

● Proposed method

- **Multimodal fusion approach:** Merge text-based large language model (LLM) predictions with image-based classifiers
- **Iterative process:** Alternately refine speaker prediction using character labels, and character identification using text labels



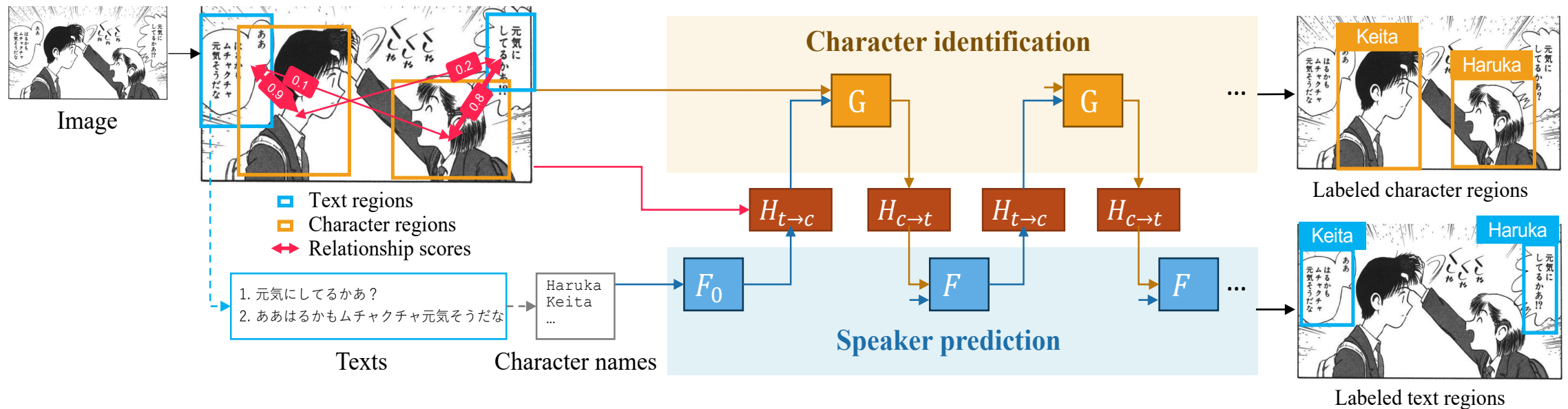
Overall framework



● Data preprocessing

- **Object detection:** Get character regions and text regions
- **Relationship prediction:** Get initial relationship scores
- **OCR:** Get texts
- **Character name extraction:** Get target labels

Overall framework



● Main pipeline: Three modules

- Speaker prediction (F):

Textual info. + Character labels $\rightarrow$ Text labels

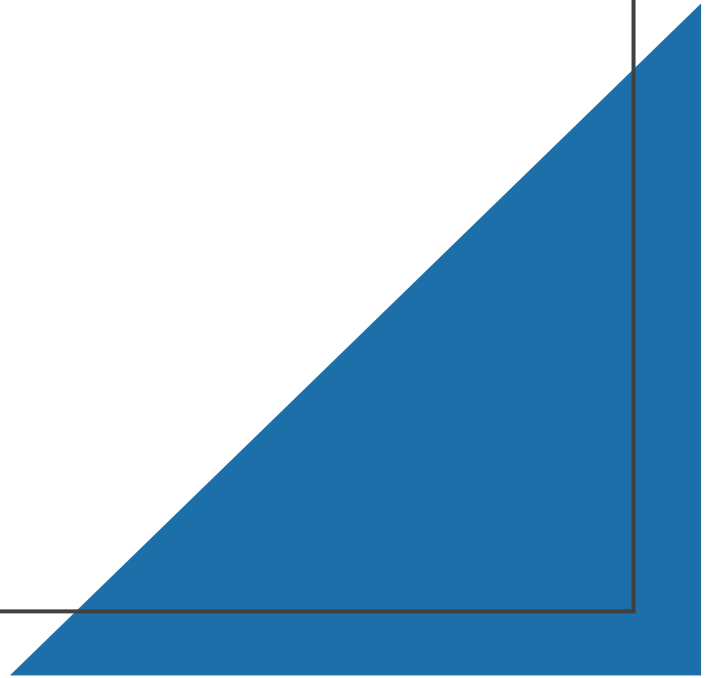
- Character identification (G):

Visual info. + Text labels $\rightarrow$ Character labels

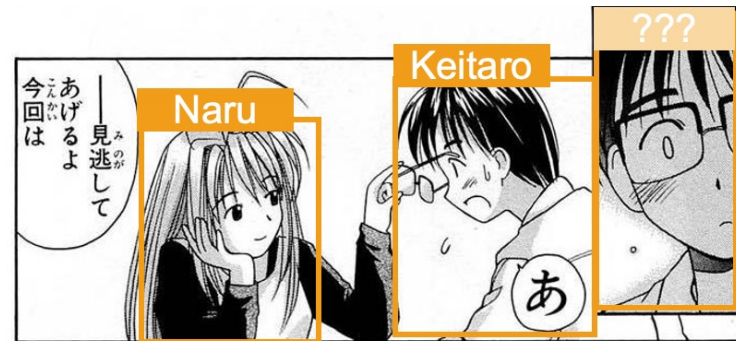
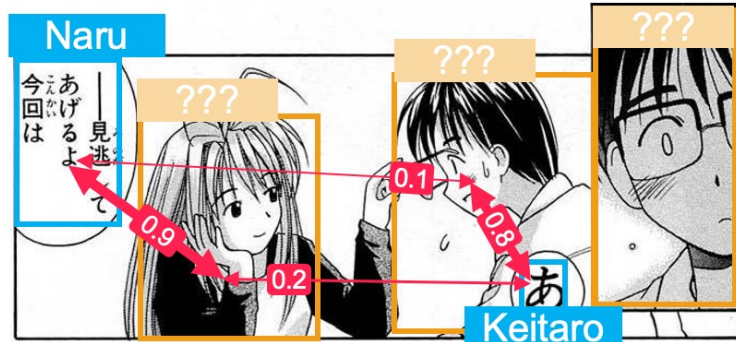
- Label propagation ($H_{t \rightarrow c}$, $H_{c \rightarrow t}$):

Text labels $\leftrightarrow$ Character labels

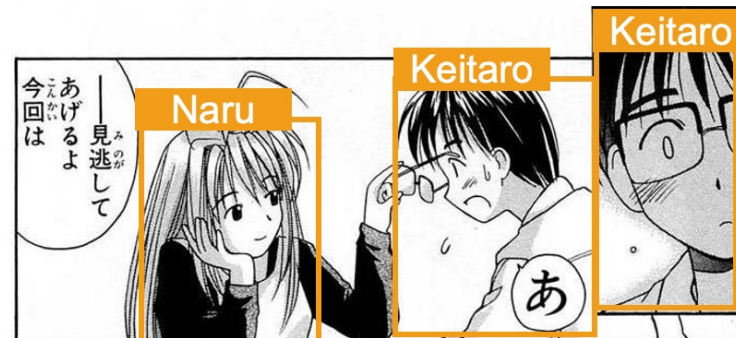
- Introduction
- **Approach**
- Experiments
- Discussion and Conclusion



Iterative character identification



Pseudo labels



Predicted labels

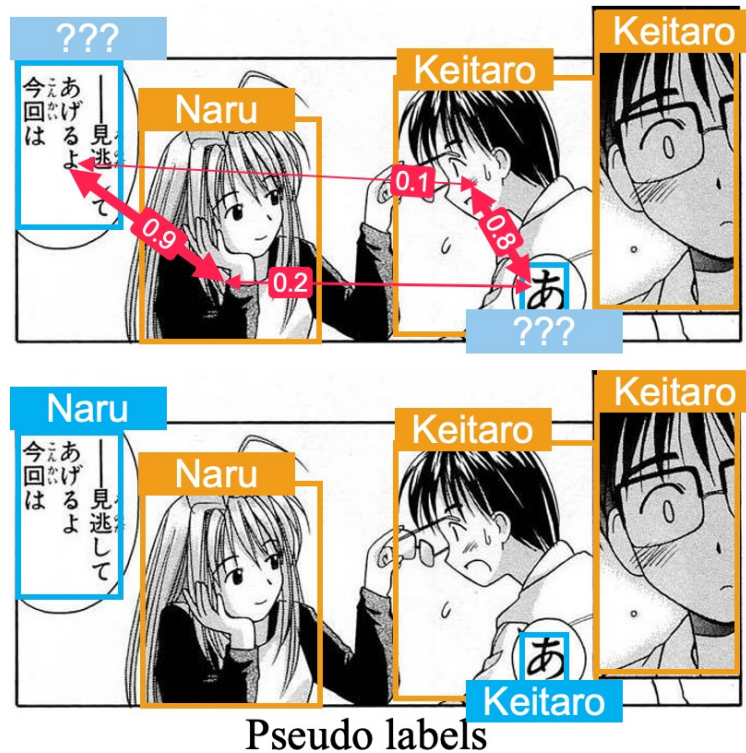
● Step 1: Pseudo label generation

- Select the character–text pair with the highest relationship score
- Take the text region's label as the pseudo label

● Step 2: Character identification

- Fine-tune a classifier with pseudo labels
- Apply the classifier to character regions

Iterative speaker prediction



- Step 1: Pseudo label generation

- Step 2: Speaker prediction

- Treat pseudo labels as candidate speakers

Input

Given dialogs and character names in comics, annotate the speaker of each dialog.

Characters:

A | Keitaro
B | Naru

Character names

Context:

Story: Keitaro tries to ...
Characters:
- Keitaro: Main character who ...
- Naru: Heroine of the story. ...

Context information (obtained beforehand by LLMs)

Dialog:

1 | Hey, Naru. | Keitaro (0.56)
2 | What, Keitaro? | Naru (0.8)

Texts Speaker candidates

Prompt

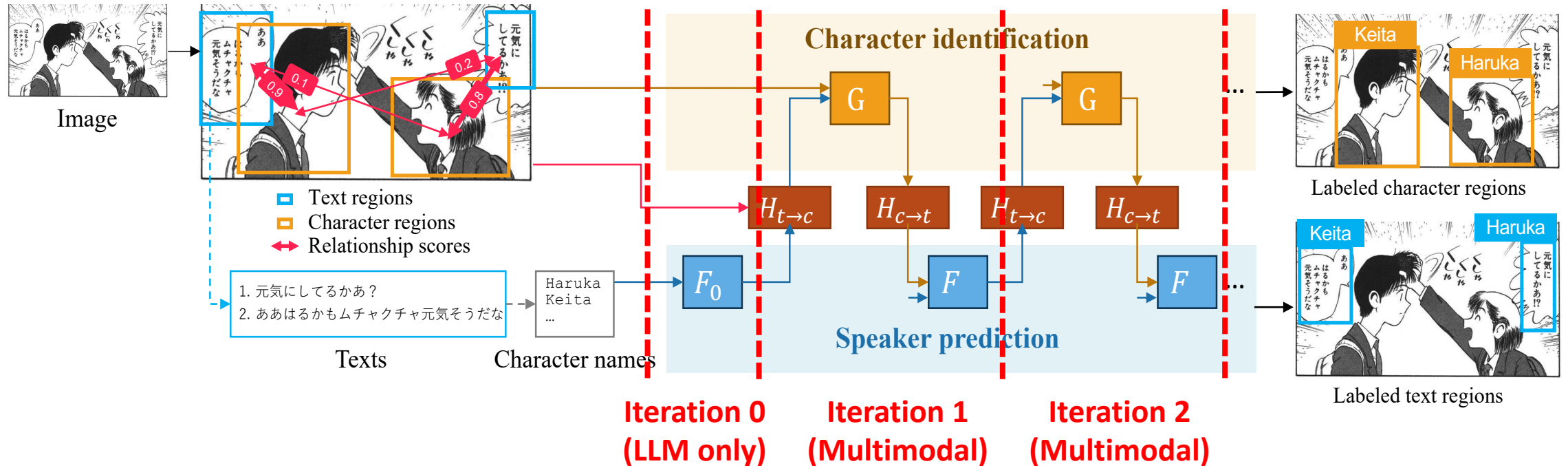
LLM

Output

L	Character	Id	Confidence
1	Keitaro	A	5
2	Naru	B	4

Predicted labels

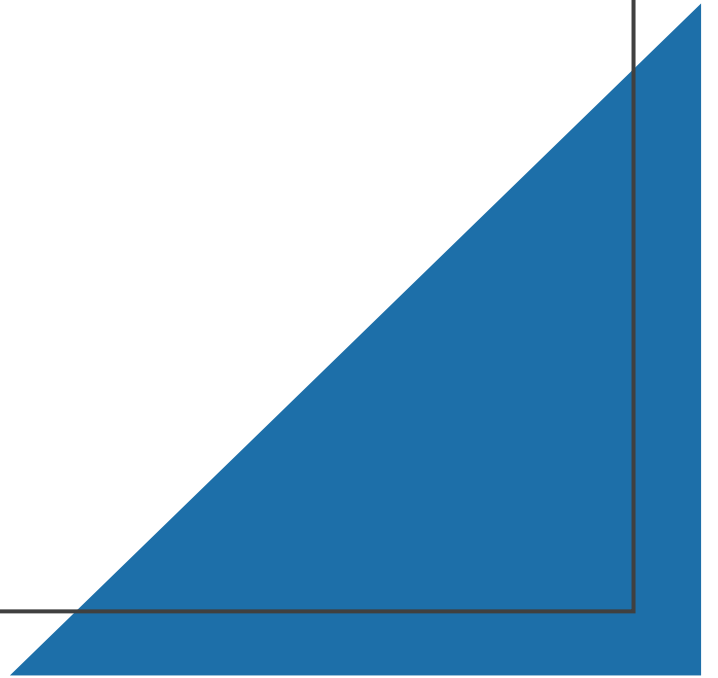
Iterative process



● Iterative performance enhancement

- Predictions from the previous step → Close to true labels
- Generated pseudo labels → Reliable training data
- Updated prediction results → High accuracy

- Introduction
- Approach
- **Experiments**
- Discussion and Conclusion



Quantitative results

- **Main results:** Simplified the tasks to classifying the character labels for the character and text regions
- **Baselines:** Clustering + mapping clusters to ground truth*
- **Data division:** Divided the test set by the difficulty of relationship prediction

	iter	text	img	Speaker pred.			Character id.		
				<i>Easy</i>	<i>Hard</i>	<i>Total</i>	<i>Easy</i>	<i>Hard</i>	<i>Total</i>
Baseline									
K-means+Distance	-		✓	34.5*	31.8*	33.1*	37.0*	36.7*	36.8*
K-means+SGG	-		✓	36.7*	34.8*	35.7*	37.0*	36.7*	36.8*
Proposed									
LLM only	0	✓		41.8	45.1	43.6	-	-	-
Multimodal	1	✓	✓	51.0	51.2	51.1	45.8	39.6	42.4
	2	✓	✓	52.4	51.3	51.8	48.5	40.3	44.0
	3	✓	✓	53.5	49.8	51.6	48.9	37.7	42.8

Our multimodal method shows a significant improvement over unimodal methods

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Our iterative process is effective on both subsets, particularly on *Easy*

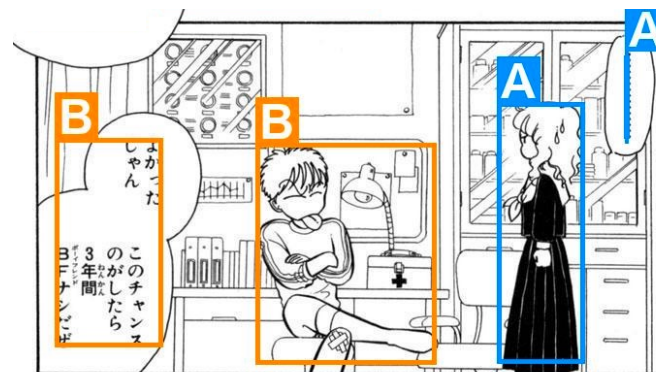
Qualitative results

● Unimodal vs. Multimodal

■ Mislabeled predictions (ground truth)



LLM only



Multimodal

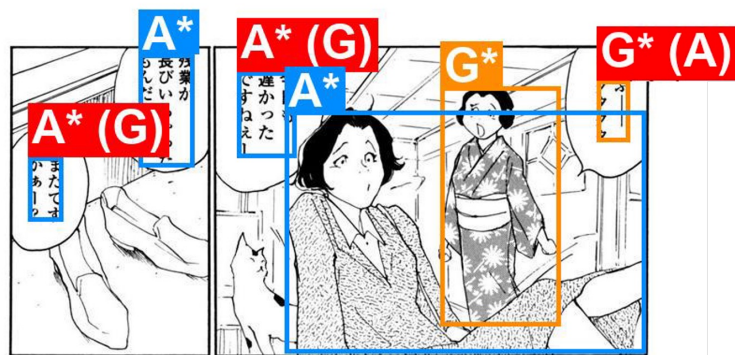
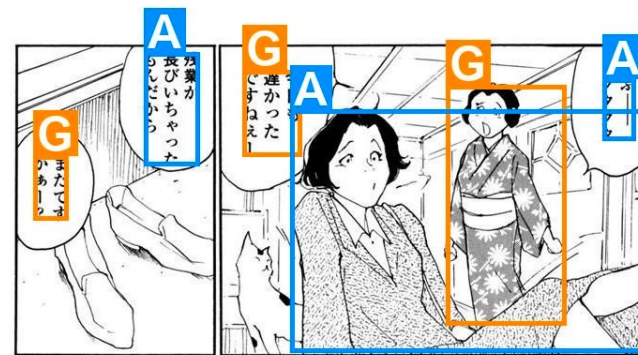


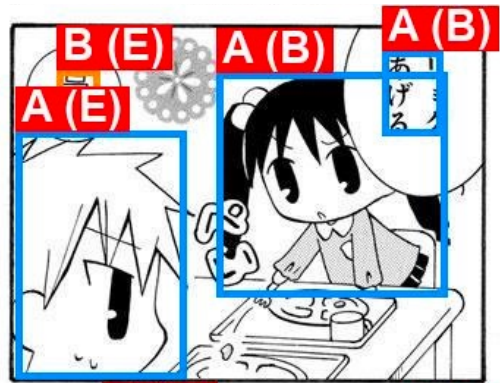
Image only (using GT*)



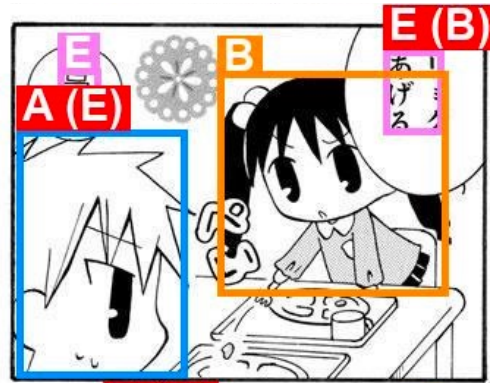
Multimodal

Qualitative results

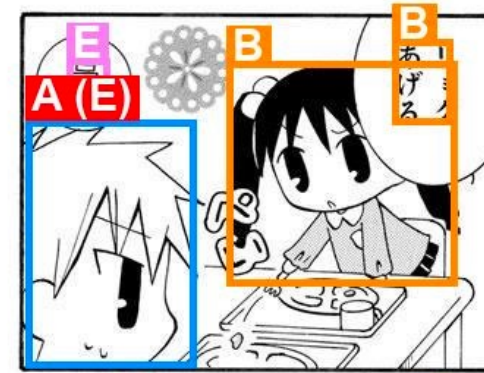
- **One-step vs. Iterative** (Accuracy of speaker pred. & character id.)



Iteration 1 (0/3 & 0/3)



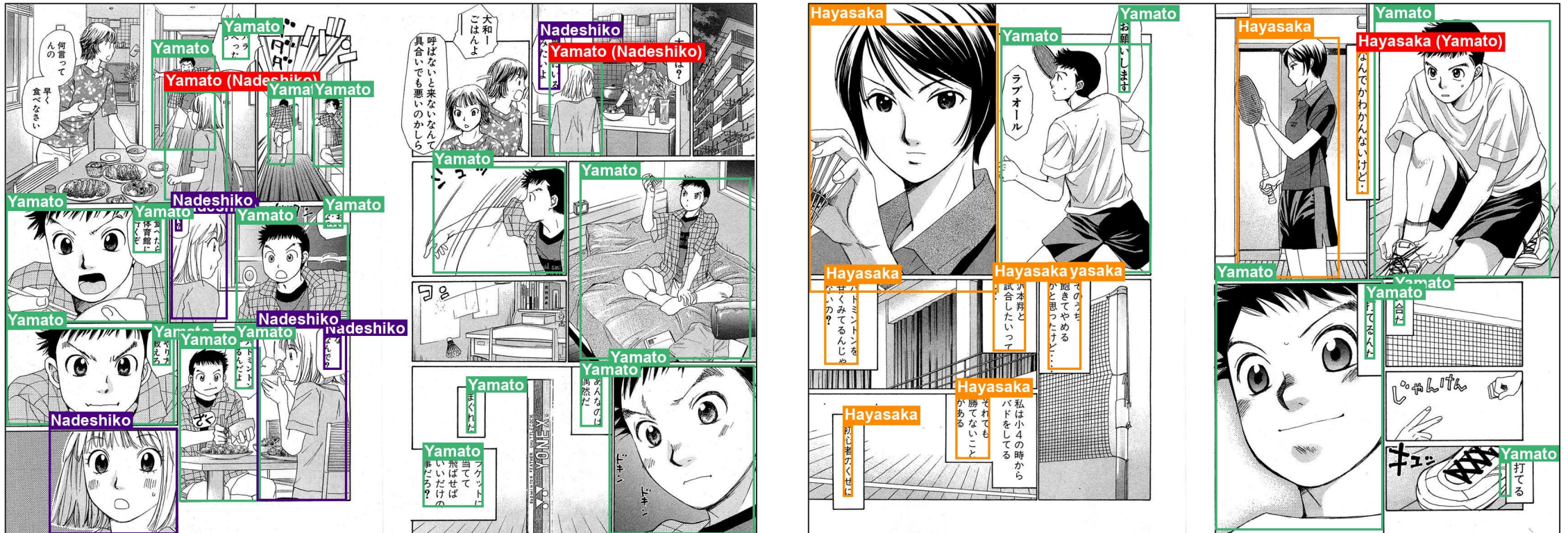
Iteration 2 (1/3 & 1/3)



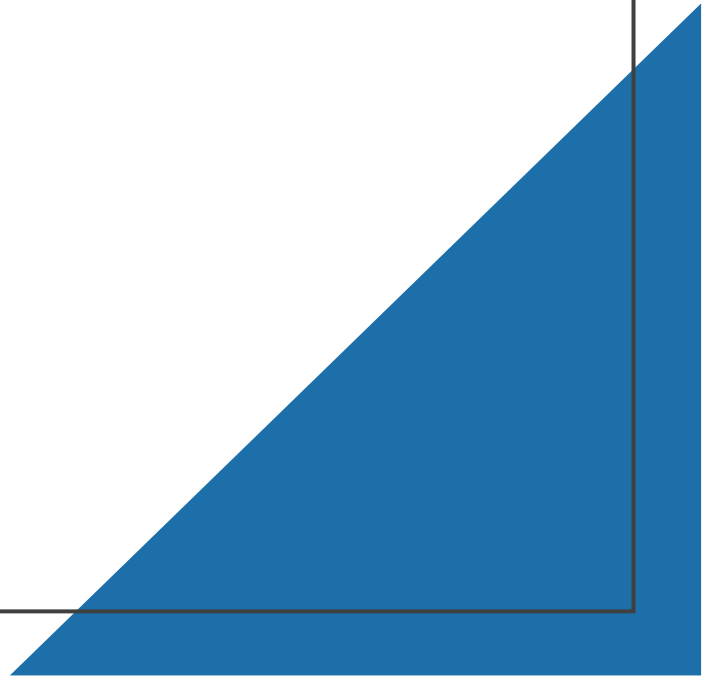
Iteration 3 (3/3 & 2/3)

Zero-shot results

- Results under entirely zero-shot settings



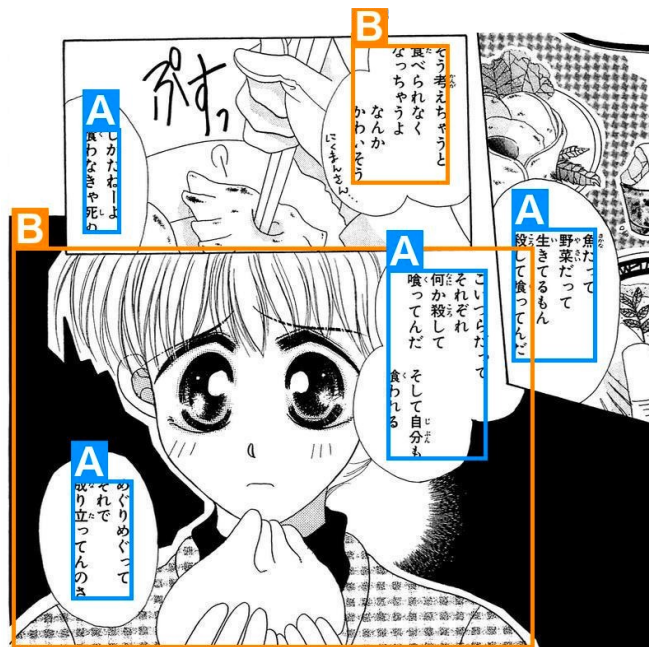
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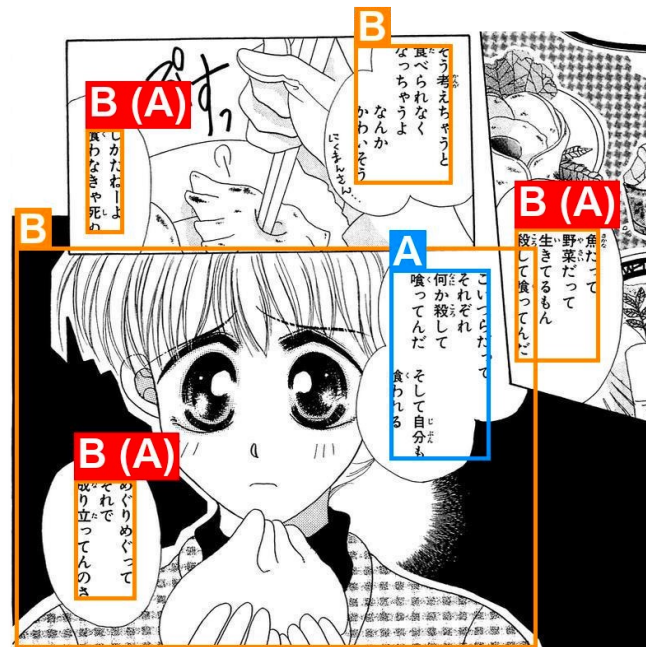
Discussion

● Limitations of proposed method

- The overall accuracy is limited
- More iterations do not necessarily lead to higher accuracy



Iteration 1



Iteration 2

Conclusion

- **New tasks**

- First to integrate the tasks of character identification and speaker prediction in comics
- First to tackle zero-shot tasks with direct applications in real-world scenarios

- **Iterative multimodal fusion**

- Revealing the significant potential of LLMs for comics analysis
- First approach to use multimodal information for character identification and speaker prediction

- **Future work**

- Refine the model and enhance accuracy

Explore more

- Visit our poster presentation!
 - **Poster session:** Poster Session 3
 - **Posterboard:** P175
- Explore our project page for more details



Thank you!

